# FIITJEE
# ALL INDIA TEST SERIES
## JEE (Advanced)-2023
## FULL TEST – V
## PAPER –2
## TEST DATE: 08-02-2023

**Time Allotted: 3 Hours** **Maximum Marks: 180**

**General Instructions:**

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

**Section – A *(01 – 04, 19 – 22, 37 – 40)*:** This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

**Section – A *(05 –10, 23 – 28, 41 – 46)*:** This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

**Section – B *(11 – 18, 29 – 36, 47 – 54)*:** This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.**

**MARKING SCHEME**

**Section – A (Single Correct):** Answer to each question will be evaluated according to the following marking scheme:

| | | | |
|---|---|---|---|
| *Full Marks* | : | +3 | If ONLY the correct option is chosen. |
| Zero Marks | : | 0 | If none of the options is chosen (i.e. the question is unanswered); |
| *Negative Marks* | : | –1 | In all other cases. |

**Section – A (One or More than One Correct):** Answer to each question will be evaluated according to the following marking scheme:

| | | | |
|---|---|---|---|
| Full Marks | : | +4 | If only (all) the correct option(s) is (are) chosen; |
| Partial Marks | : | +3 | If all the four options are correct but ONLY three options are chosen; |
| Partial marks | : | +2 | if three or more options are correct but ONLY two options are chosen and both of which are correct; |
| Partial Marks | : | +1 | If two or more options are correct but ONLY one option is chosen and it is a correct option; |
| Zero Marks | : | 0 | If none of the options is chosen (i.e. the question is unanswered); |
| Negative Marks | : | –2 | In all other cases. |

**Section – B:** Answer to each question will be evaluated according to the following marking scheme:

| | | | |
|---|---|---|---|
| Full Marks | : | +3 | If ONLY the correct integer is entered; |
| Zero Marks | : | 0 | Question is unanswered; |
| Negative Marks | : | –1 | In all other cases. |

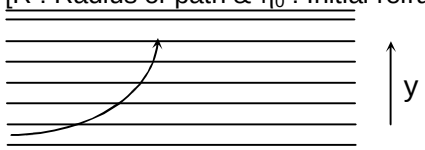
# Physics

## PART – I

### Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

1. The refractive index of the medium within a certain region, x > 0, y > 0, changes with y. A thin light ray travelling in the x-direction strikes the medium at right angles and moves through the medium along a circular arc.
How does the refractive index depend on y?
[R : Radius of path & $\eta_0$ : Initial refractive index]

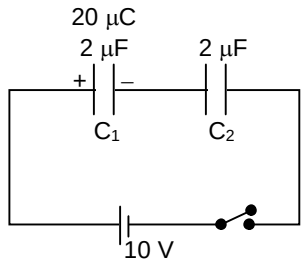



(A) $\frac{2R}{R-y}\eta_0$ (B) $\frac{R}{R-y}\frac{\eta_0}{2}$

(C) $\frac{R+y}{R}\eta_0$ (D) $\frac{R}{R-y}\eta_0$

2. A mass is suspended from a wire and is pulled along the length of wire, resulting in oscillations of time period $T_1$. The same mass is next attached to wire of the same material and length but double the cross-sectional area. The time period this time is $T_2$. Then $T_1/T_2$ is equal to
(A) 1 : 2 (B) 2 : 1
(C) 1 : $\sqrt{2}$ (D) $\sqrt{2}$ : 1

3. Initially, $C_1$ is charged but $C_2$ is uncharged. When switch is closed, charges on $C_1$ and $C_2$ are
(A) 20 μC, 0
(B) 10 μC, 10 μC
(C) 30 μC, 10 μC
(D) 0, 20 μC




4. A standing wave is produced in a vapour of an element of atomic mass 254 (g $mol^{-1}$) at 400 K. Nodes are found to be 6.77 cm apart when frequency of source was 1000 Hz. The vapour used for the experiment is (R = 8.31 $JK^{-1}mol^{-1}$)
(A) monatomic (B) diatomic
(C) polyatomic (D) none of these

## Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

5. A particle of mass $m$ is moving in a field where the potential energy is given by $U(x) = U_0(1-\cos ax)$, where $U_0$ and $a$ are constants and $x$ is the displacement from mean position . Then (for small oscillations)

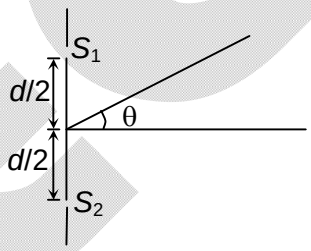
(A) the time period is $T = 2\pi\sqrt{\frac{m}{aU_0}}$

(B) the speed of particle is maximum at $x = 0$

(C) the amplitude of oscillations is $\frac{\pi}{a}$

(D) the time period is $T = 2\pi\sqrt{\frac{m}{a^2U_0}}$

6. In an interference arrangement similar to Young's double slit experiment, the slits $S_1$ and $S_2$ are illuminated with coherent microwave sources, each of frequency $10^6$Hz. The sources are synchronized to have zero phase difference. The slits are separated by a distance d = 150.0 m. The intensity $I(\theta)$ is measured as a function of $\theta$, where $\theta$ is defined as shown. If $I_0$ is maximum intensity, then $I(\theta)$ for $0 \le \theta \le 90°$ is given by




(A) $I(\theta) = \frac{I_0}{2}$ for $\theta = 30°$

(B) $I(\theta) = \frac{I_0}{4}$ for $\theta = 90°$

(C) $I(\theta) = I_0$ for $\theta = 0°$

(D) $I(\theta)$ is constant for all values of $\theta$

7. A uniform square plate of mass m and edge a initially at rest starts rotating about one of the edge under the action of a constant torque $\tau$. Then at the end of the 5th sec after start

(A) angular momentum is equal to $5\tau$

(B) kinetic energy is equal to $\frac{75\tau^2}{ma^2}$

(C) angular momentum is equal to $2.5\tau$

(D) kinetic energy is equal to $\frac{75\tau^2}{2ma^2}$

8. A particle of mass $m$ start moving at $t = 0$ due to a force $F = F_0\cos\omega t$, where $F_0$ and $\omega$ are constant. Particle will stop first time at $t = t_0$ and distance travel in this time interval is $S$, then

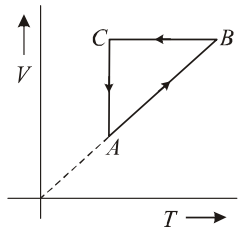
(A) $T_0 = \frac{\pi}{\omega}$

(B) $t_0 = \frac{\pi}{2\omega}$

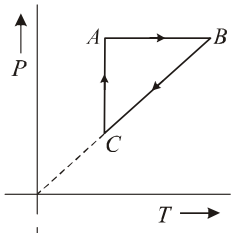
(C) $S = \frac{2F_0}{m\omega^2}$

(D) $S = \frac{F_0}{m\omega^2}$

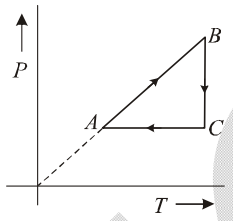
9. An ideal gas under goes a thermodynamic cycle as shown in figure. Which of the following graphs represents the same cycle ?



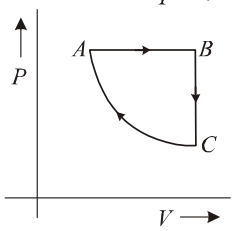

(A)



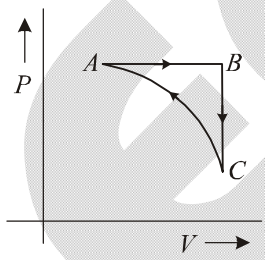

(B)



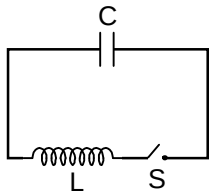

(C)




(D)




10. In the given circuit, initially the charge on capacitor is $Q_0$. At t = 0, the switch S is closed. Which of the following statement(s) is/are correct?




(A) Maximum current through the inductor is $\frac{Q_0}{\sqrt{LC}}$. b

(B) Charge on the capacitor is zero at $t = \frac{\pi}{2}\sqrt{LC}$.

(C) Current through the inductor is maximum at $t = \frac{\pi}{2}\sqrt{LC}$.

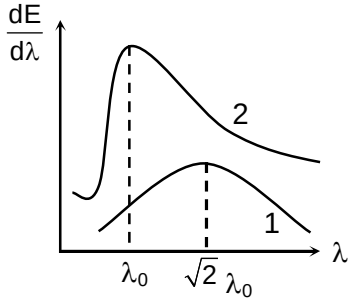
(D) Maximum magnetic energy can be stored in the inductor is $\frac{Q_0^2}{2C}$.

## Section – B (Maximum Marks: 24)

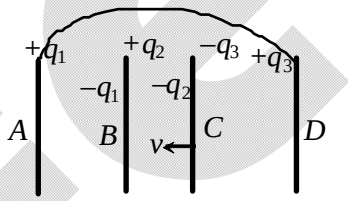
*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

11. A vertical hollow cylinder contains an ideal gas. The gas is enclosed by a 5 kg movable piston with area of cross-section $5 \times 10^{-3}$ $m^2$. The gas is slowly heated from 300 K to 350 K and the piston rises by 0.1 m. The piston is now clamped at this position and the gas is cooled back to 300 K. The difference between the heat energy added during heating process and heat energy lost during cooling process is 11*x*. Find value of *x*. [Take $p_0 = 10^5$ $N/m^2$]

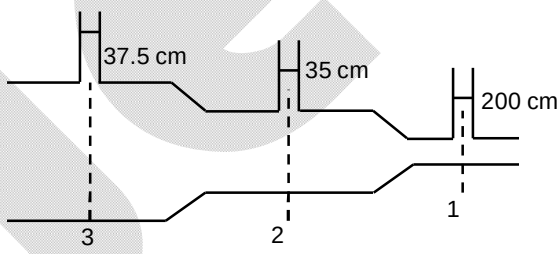
12. Graph 1 and 2 represent the spectral emissive power of a body at temperature $T_1$ and $T_1$. The ratio of area under curve 2 and curve 1 is.




13. In the diagram there are four conducting plates *A*, *B*, *C* and *D* placed parallel to each other at equal separation *L*. If plate *C* starts moving towards plate *B* with velocity *v*. Find the current (in mA) flowing in the wire connecting *A* and *D*. (assume all other plates to be fixed)
(Given: $q_2 = 2\mu C$, $q_3 = 3\mu C$, $v = 30$ m/s, $L = 0.05$ m)



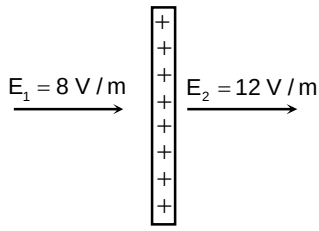

14. Water is flowing in varying cross-section pipe. The areas of cross-sections 1, 2 and 3 are 1 $cm^2$, 2 $cm^2$ and A $cm^2$, respectively. Water levels are shown in different vertical tubes of manometers. The speed of water at cross-section 3 is $\frac{1}{\sqrt{x}}$ m/s.
The value of x is




15. If the $K_\alpha$ radiation of Mo (Z = 42) has a wavelength of 0.71 Å, calculate wavelength of the corresponding radiation of Cu. i.e. $K_\alpha$ for Cu (Z = 29). Find the value of nearest integer value (in Å) of wavelength.

16. A particle is suspended by a light vertical inelastic string of length $\ell$ from a fixed support. At its equilibrium position, it is projected horizontally with a speed $\sqrt{6g\ell}$. What is the ratio of the tension in the string in its horizontal position to that in string, when the particle is vertically above the point of support?

17. The electric field on two sides of a charged plate is shown in the figure. If the charge density on the plate is $n\varepsilon_0$, then n is




18. A hydrogen like atom (atomic number z) is in a higher excited state of quantum number n. The excited state by successively emitting two photons of energy 10.2 eV and 17.0 eV, respectively. Alternative, the atom from the same excited by successively emitting two photons of energies 4.25 eV and 5.95 eV, respectively. Determine the value of n. (Take, ionisation energy of H-atom = 13.6 eV)

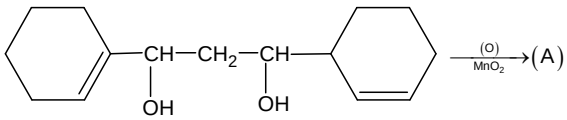
# Chemistry

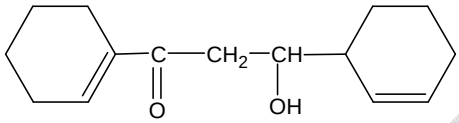
## PART – II

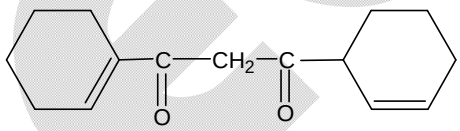
### Section – A (Maximum Marks: 12)

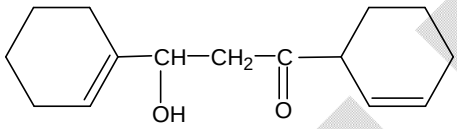
*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

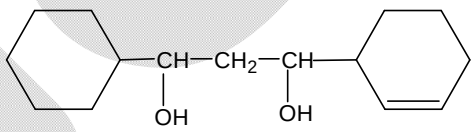
19. For $[HgI_3]^-$ ion geometry and magnetic behaviour is respectively:
(A) triangular bipyramideal, paramagnetic
(B) triangular pyramidal, diamagnetic
(C) triangular planar, paramagnetic
(D) triangular planar, diamagnetic

20. 

The product (A) is
(A) 

(B) 

(C) 

(D) 


21. For most reactions, rate of a reaction increases about two times when its temperature is raised by 10 K. This fact is due to
(A) increase in number of collisions per unit volume
(B) increase in number of molecules having energy $E_A$ or more
(C) lowering of activation energy
(D) lowering of pre-exponential factor in Arrhenius equation

22. At constant volume, for a fixed amount of a gas the pressure of the gas increase with rise of temperature due to
(A) increase in average molecular speed
(B) increased rate of collision amongst molecules
(C) increase in molecular attraction
(D) decrease in mean free path

### Section – A (Maximum Marks: 24)

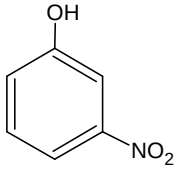
*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

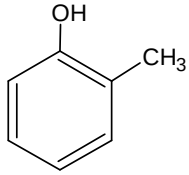
23. Which of the following statements are correct?
(A) Copper metal can reduce $Fe^{2+}$ in acidic medium.
(B) The absolute value of standard electrode potential can be determined experimentally
(C) In a Daniell cell, electrons flow from zinc electrode to copper electrode outside the cell
(D) Cell voltage is an intensive quantity.

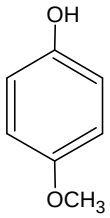
24. Which of the following show the correct order of nucleophilicity?

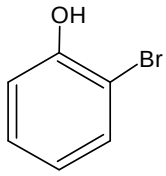
(A) $CH_3S^- > CH_3O^-$
(B) $(CH_3)_3 N > (CH_3)_3 \ddot{P}$
(C) $CN^- > Cl^-$
(D) $\overline{O}C_2H_5 > \overline{O}H$

25. Which of the following is/are more acidic than phenol?

(A) 


(B) 


(C) 


(D) 

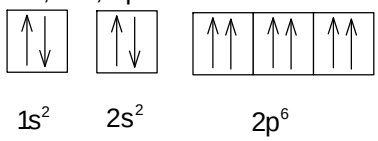

26. Select molecules(s) in which all the $\pi$-orbitals are in same plane.

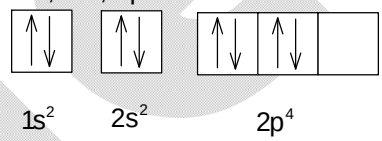
(A) $C_6H_6$ (B) $O = C = C = C = O$

(C) $H_2C = C = CH_2$ (D) $CH_2 = CH - CH = CH_2$

27. In which of the following Pauli rule is violated?

(A) $1s^2, 2s^2, 2p^7$ (B) $1s^1, 2s^1, 2p^6$

(C) 


(D) 


28. Synergic bonding in metal carbonyl complexes [M = Metal]:

(A) decreases C – O bond strength

(B) increases M – C bond length

(C) increases bond order for M – C bond

(D) increases paramagnetism of complex

## Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

29. Find the total number of electron in Cr which have $(n \times l) =$ zero

30. Total number of exchanges possible for $3d^4$ electronic configuration are?

31. Total number of bonds in $PCl_5$ which are at $90^o$ to each other, are

32. Consider octahydral complex $K_2$[Ni(EDTA)].
Total no. of 5 member rings in compound = X
Total number of geometrical isomers possible for compound = Y
Represent your answer as a total of X and Y.

33. Consider a titration of potassium dichromate solution with acidified Mohr's salt solution using diphenylamine as indicator. The amount of Mohr's salt required per mole of dichromate is

34. The number of atoms in a single face-centred cubic unit cell is

35. Total no. of geometrical isomers possible for 1, 2, 4-tribormocyclohexane are?

36. Total no. of stereoisomers possible for 1, 2-dimethylcyclopropane are?

# Mathematics

## PART – III

### Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

37. If the area bounded by $f(x) = \sqrt{1+x^2}(x > 0)$ the line y = x, y axis and y = –x + a(a>1) is k, then area bounded by the graph of $f^{-1}(x)$, the line y = x between the lines y = –x + 1 and y = –x + a is

(A) $\frac{k-1}{4}$ (B) $\frac{2k-1}{2}$

(C) k (D) $\frac{4k-1}{4}$

38. In a acute angle triangle ABC, the foot of the perpendicular from A divide the opposite side into two parts of length 3 and 17 units and $\tan A = \frac{22}{7}$, the $\left(\frac{\text{Area of } \Delta ABC}{22}\right)$ is

(A) $\frac{5}{2}$ (B) 5

(C) 10 (D) 20

39. The digits of three digit number N are in A.P. If sum of the digits is 15 and the number obtained by reversing the digits of the number is 594 less the original number, then $\frac{1000}{N-252}$ is equal to :

(A) $\frac{5}{6}$ (B) $\frac{5}{3}$

(C) 0.06 (D) 0.03

40. A flagstaff stands vertically on a pillar, the height of the flagstaff being double the height of the pillar. A man on the ground at a distance finds that both the pillar and the flagstaff subtend equal angles at his eyes. The ratio of the height of the pillar and the distance of the man from the pillar, is

(A) $\sqrt{3}:1$ (B) 1 : 3

(C) $1:\sqrt{3}$ (D) $\sqrt{3}:2$

### Section – A (Maximum Marks: 24)

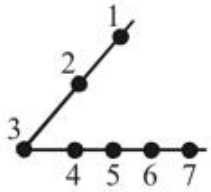
*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

41. Let $\vec{a},\vec{b},\vec{c}$ be noncoplanar vectors and $\vec{p} = \frac{\vec{b}\times\vec{c}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}$, $\vec{q} = \frac{\vec{c}\times\vec{a}}{\left[\vec{b}\,\vec{c}\,\vec{a}\right]}$, $\vec{r} = \frac{\vec{a}\times\vec{b}}{\left[\vec{c}\,\vec{a}\,\vec{b}\right]}$ then

(A) $\vec{p}\cdot\vec{a} = 1$ (B) $\vec{p}\cdot\vec{a}+\vec{q}\cdot\vec{b}+\vec{r}\cdot\vec{c} = 3$

(C) $\vec{p}\cdot\vec{a}+\vec{q}\cdot\vec{b}+\vec{r}\cdot\vec{c} = 0$ (D) none of these

42. 7 points are given in a plane (as shown in adjacent figure), then using these 7 points –




(A) Number of lines that can be formed is 10.
(B) Number of triangles that can be formed is 24.
(C) Number of quadrilateral that can be formed is 6.
(D) Maximum number of circles that can be formed is 24, where each circle contains atleast three of 7 points

43. The equations of four circles are $(x \pm a)^2 + (y \pm a)^2 = a^2$. The radius of a circle touching all the four circles is

(A) $\left(\sqrt{2}-1\right)a$ (B) $2\sqrt{2}a$

(C) $\left(\sqrt{2}+1\right)a$ (D) $\left(2+\sqrt{2}\right)a$

44. If f(x) is a cubic polynomial such that f(0) + 2 = 0 and $f'''(0) = 6 \;\&\; f(x) = 0$ posses three positive integral roots then

(A) $f'(1) = 0$ (B) $2f'(0) + f''(0) = 0$

(C) $2f'(0) + f''(0) = 2$ (D) $f'(2) = 0$

45. If $x + [y] + \{z\} = 1.1$ [.] is G.I.F and {.} is fractional part
$[x] + \{y\} + z = 2.2$
$\{x\} + y + [z] = 3.3$ then

(A) $x + y + z = 3.3$ (B) $y - 2x = 1$

(C) $2(z+1) = 5y$ (D) $\{x\} + \{y\} + \{z\} = 0.3$

46. If the sides a, b & c of a triangle ABC are in G.P. then $\angle B$ can never be

(A) $\frac{\pi}{6}$ (B) $\frac{5\pi}{6}$

(C) $\frac{\pi}{4}$ (D) $\frac{2\pi}{3}$

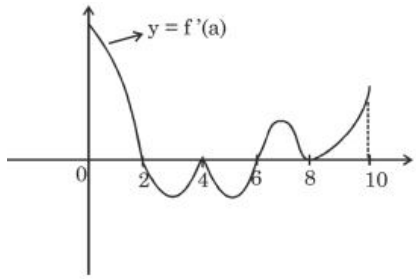
## Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

47. If $S = 3.\sqrt{9}.\sqrt{\sqrt{27}}.\sqrt{\sqrt{\sqrt{81}}}....\infty$, then sum of digits in S is

48. Let f(x) = (1 + x) $\ell n(1+x) - x - \frac{x^2}{4}$, $\forall x > -1$ and $f'(x) > 0 \forall \in x(0, \alpha)$, then value of $[\alpha]$ is (where denotes greatest integer function)

49. Let f(x) be a derivable function such that f(0) = 1, f(2) = 2, f(4) = 1 and f(6) = – 4. The graph of y $y = f'(x)$ is given below. Then find the total number of points of local maximum/local minimum of y =f (x) ln (0, 10)




50. The projection length of a variable vector $x\hat{i}+y\hat{j}+z\hat{k}$ on the vector $\vec{p}=\hat{i}+2\hat{j}+3\hat{k}$ is 6. Let $\ell$ be the minimum projection length of the vector $x^2\hat{i}+y^2\hat{j}+z^2\hat{k}$ on the vector $\vec{p}$, then the value of $\sqrt[3]{l^2+15^2}$ is

51. Non-zero vectors $\vec{a},\vec{b},\vec{c}$ satisfy $\vec{a}.\vec{b}$ = 0, $(\vec{b}-\vec{a}).(\vec{b}+\vec{c})$ = 0 and $2|\vec{b}+\vec{c}|=|\vec{b}-\vec{a}|$. If $\vec{a}=\mu\vec{b}+4\vec{c}$ then the value of $\mu$ is

52. 3 couples have to be seated around a circle. Let p be the probability that no couple is together then the value of 30 p is

53. If $f(0)=1, f(2)=3, f'(2)=5$ then the value of the definite integral $\int_0^1 x f''(2x)dx$ is

54. A special die is so constructed that the probabilities of throwing 1, 2, 3, 4, 5 and 6 are (1 – k)/6, (1 + 2k)/6, (1 – k)/6, (1 + k)/6, (1 – 2k)/6 and (1 + k)/6 respectively. If two such dice are thrown and the probability of getting a sum equal to 9 lies in $\left[\frac{1}{9},\frac{2}{9}\right]$. Then find the number of integral solutions of k.